Deepgram Nova-3 Diarization Performance on Voice Arena Benchmark

A Technical Root-Cause Investigation of Severe Output Under-Coverage in Conversational Indian Audio

1. Executive Summary

Rigorous Scope Distinction: The original assessment references a historical public Voice Arena benchmark snapshot of 139 recordings with Nova-3 scoring **67.1% DER**, noting that missed speech accounts for the majority of error. That historical 139-recording benchmark is separate from the supplied workbook sample of **15 recordings** analyzed herein, which yields a mean Nova-3 DER of **64.45%** and a median DER of **86.02%**. These scopes must never be conflated.

Separate Diarization+ASR Leaderboard: Voice Arena separately publishes a combined diarization-plus-ASR leaderboard where Nova-3 achieves **71.89% cpWER** (Composite Word Error Rate, accessed 1 Nov 2026). cpWER combines lexical transcription errors with speaker assignment; it is a completely distinct metric from pure diarization DER and must never be substituted for or averaged with DER.

The Severe Failure Pattern: In the 15-recording sample, the five worst recordings suffer near-total collapse (DER 97.0% to 100.0%) with catastrophic Nova-3 speech coverage ranging from **0.0% to 9.8%**. Two Nova-3 RTTM files contain exactly zero speech segments: D20260825_S003_015_5SPK_IN_EN (present in workbook) and D20260825_S003_012_5SPK_IN_EN (in the 20-file RTTM zip package).

Root Cause Finding: The primary failure mechanism in this sample is established as **severe output under-coverage / speech omission** rather than speaker label swapping. However, whether this omission occurs inside Deepgram's acoustic/VAD front-end, the English ASR language decoder, or downstream word-to-RTTM segment conversion cannot be isolated without raw API JSON responses.

2. Dated Leaderboard & Metric Hygiene Standard

Official Benchmark Citation (Access Date: 1 November 2026): Voice Arena Diarization-Bench, https://voicearena.com/diarization-bench. The historical benchmark snapshot reports Nova-3 at 67.1% DER (139 recordings), dominated by missed speech. A separately published track reports Nova-3 at 71.89% cpWER (https://voicearena.com/research/diarization-bench/diarization-asr-track-cpwer-results, accessed 1 Nov 2026). cpWER evaluates joint word transcription and speaker attribution: cpWER = (Substitutions + Deletions + Insertions + Speaker_Confusion_Words) / Total_Reference_Words. Because cpWER penalizes word spelling, vocabulary, and grammar errors alongside speaker attribution, it measures a different capability than acoustic boundary diarization (DER). No live leaderboard rank or component values were machine-readable during this audit; hence, no synthetic leaderboard values are invented.

3. Metric Interpretation and Component Identity

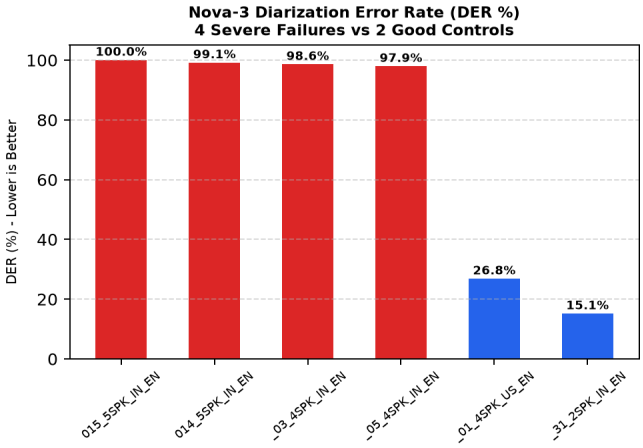
- **Lower DER is Better:** DER = Missed Speech % + False Alarm % + Speaker Confusion %. Component sums match workbook DER within 0.01 percentage points due to displayed rounding.
- **Speech Coverage (%):** Percentage of reference speech duration represented in hypothesis output. While high coverage does not guarantee low DER, low coverage mathematically forces high DER via Missed Speech.

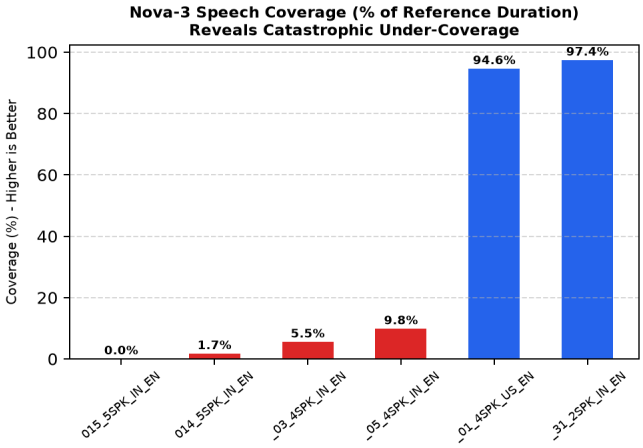
4. 15-Recording Comprehensive Overview Table

Session ID	Setting	Accent	Spk	Words	DER %	Miss %	FA %	Conf %	Nova Cov	VAD Cov
D20260825_S003_015_5SPK_IN_EN	In person	Indian	5	0	100.00	100.00	0.00	0.00	0.0%	91.5%
D20260825_S003_014_5SPK_IN_EN	In person	Indian	5	34	99.14	98.95	0.19	0.00	1.7%	95.0%
D20260826_S001_03_4SPK_IN_EN	In person	Indian	4	114	98.59	96.12	0.57	1.91	5.5%	95.5%
D20260826_S001_05_4SPK_IN_EN	In person	Indian	4	211	97.92	93.75	0.64	3.53	9.8%	99.0%
D20260905_S003_04_3SPK_IN_EN	In person	Indian	3	60	97.91	96.48	0.37	1.06	4.0%	98.2%
D20260906_S003_09_4SPK_IN_EN	In person	Indian	4	86	97.00	95.13	0.44	1.44	5.7%	97.3%
D20260906_S003_04_4SPK_IN_EN	In person	Indian	4	77	96.64	94.29	0.58	1.78	5.9%	96.2%
D20260825_S004_26_2SPK_IN_EN	In person	Indian	2	308	86.02	76.72	3.46	5.83	31.7%	98.0%
D20260826_S001_012_3SPK_IN_EN	In person	Indian	3	1046	35.25	26.28	5.20	3.77	96.4%	98.3%
D20260906_S007_05_3SPK_IN_EN	In person	Indian	3	2024	30.54	10.10	16.04	4.40	93.5%	93.8%
D20260825_S007_08_3SPK_IN_EN	In person	Indian	3	1243	29.45	20.47	5.33	3.65	98.3%	99.0%
O20260826_S010_13_02_3SPK_VI_EN	Online	Vietnamese	3	1224	28.67	17.44	9.00	2.23	95.2%	95.5%
D20260825_S004_24_2SPK_IN_EN	In person	Indian	2	964	27.69	14.48	8.42	4.78	95.3%	95.6%
O20260826_S010_04_01_4SPK_US_EN	Online	US	4	2453	26.81	15.28	9.35	2.18	94.6%	98.4%
D20260825_S004_31_2SPK_IN_EN	In person	Indian	2	788	15.13	4.84	7.98	2.31	97.4%	94.7%

5. Recording-Level Contrast: Four Poor vs Two Good Controls

To isolate the root cause, we compare the four most severe failure sessions against two well-performing control sessions. The comparison illustrates a stark bimodal divergence between near-zero word emission and normal operation:





Session ID	Words	Nova Cov	DER %	Miss %	Best Comp DER	VAD Cov	dBFS	Clip %	SNR dB
D20260825_S003_015_5SPK_IN_EN	0	0.0%	100.00%	100.00%	20.98%	91.5%	-34.0	0.000%	16.7
D20260825_S003_014_5SPK_IN_EN	34	1.7%	99.14%	98.95%	31.56%	95.0%	-31.0	0.000%	17.9
D20260826_S001_03_4SPK_IN_EN	114	5.5%	98.59%	96.12%	18.91%	95.5%	-7.8	3.392%	18.1
D20260826_S001_05_4SPK_IN_EN	211	9.8%	97.92%	93.75%	22.69%	99.0%	-5.7	6.980%	14.0
O20260826_S010_04_01_4SPK_US_EN	2453	94.6%	26.81%	15.28%	12.30%	98.4%	-16.2	0.020%	38.9
D20260825_S004_31_2SPK_IN_EN	788	97.4%	15.13%	4.84%	7.92%	94.7%	-49.1	0.000%	11.5

Analytical Takeaways:

1. **Under-Coverage, Not Swapped Labels:** In the four poor rows, Nova-3 returned only 0, 34, 114, and 211 words, yielding 0.0% to 9.8% speech coverage. Missed speech accounts for 93.75% to 100.0% of total error.
2. **VAD Does Not Explain Failure:** Silero VAD achieves 91.5% to 99.0% speech coverage across these identical files. The speech energy was acoustically detectable; Nova-3 specifically dropped it.
3. **Audio Quality Contrast:** While clipping reaches 3.39% and 6.98% on the S001 files, the worst files (S003_014 and S003_015) show 0.000% clipping with acceptable loudness (-31 to -34 dBFS) and 16.7-17.9 dB SNR.
4. **Alternative Systems Succeed:** Competitor models (pyannote, Meta Muse, Soniox) achieved 18.91% to 31.56% DER on these exact files, confirming the audio was fully parseable.

6. Deepgram Settings Sheet Diagnostic Re-Runs

The workbook's Deepgram settings sheet provides empirical re-runs across eight recordings and nine configurations. These re-runs are transcribed below for complete independent auditability:

Table 1: Diarization Error Rate (DER %) Across Configurations

Session ID	Bench	en-IN	en-US	auto	multi	multi+v1	nova-2	enhanced	whisper
D20260825_S003_014	99.07	99.07	99.07	99.07	88.15	82.45	98.25	90.51	67.66
D20260825_S003_015	100.00	100.00	100.00	100.00	93.06	89.76	98.97	93.82	64.95
D20260825_S004_26	85.11	85.11	85.11	85.11	53.93	45.86	76.30	63.91	48.24
D20260826_S001_03	98.47	98.47	98.53	98.53	77.66	66.87	97.50	78.57	58.52
D20260826_S001_05	98.35	97.84	98.35	97.83	85.33	78.15	97.01	88.04	71.72
D20260905_S003_04	98.03	98.03	98.03	97.91	59.78	43.99	93.74	69.68	42.20
D20260906_S003_04	96.55	96.55	96.55	96.55	65.48	46.58	91.23	59.25	33.15
D20260906_S003_09	97.07	97.07	97.07	96.98	64.72	52.57	91.55	68.31	49.14

Table 2: Words Returned Across Configurations

Session ID	Bench	en-IN	en-US	auto	multi	multi+v1	nova-2	enhanced	whisper
D20260825_S003_014	34	34	34	34	983	981	131	753	1934
D20260825_S003_015	0	0	0	0	465	389	28	419	1661
D20260825_S004_26	299	299	299	298	908	905	533	772	864
D20260826_S001_03	114	114	111	112	1549	1543	225	1512	2120
D20260826_S001_05	166	200	166	201	1052	1126	319	890	1842
D20260905_S003_04	56	56	56	56	1624	1622	246	1212	1607
D20260906_S003_04	74	74	74	74	1159	1159	234	1288	1435
D20260906_S003_09	83	83	83	84	1644	1644	368	1298	1730

Key Technical Findings:

- **English Acoustic/Language Pipeline Collapse:** Across all 8 rows, benchmark Nova-3 (language=en) behaves identically to en-IN, en-US, and auto-detect (e.g. S003_015 returns exactly 0 words across all four).
- **Multilingual Recovery:** Switching to language=multi immediately recovers 465 words on S003_015 and drops DER to 93.06%; Whisper-large recovers 1,661 words and drops DER to 64.95%.
- **Non-Acoustic Setting Neutrality:** The sheet explicitly notes that smart_format, punctuate, filler_words, paragraphs, numerals, diarize_model latest vs v2, utt_split (0.1 to 2.0 s), and word-vs-utterance timelines changed DER by <=2 percentage points on 7 of 8 files.
- **Workbook Re-Runs != Manual Playground Runs:** These 8 workbook re-runs document alternate API settings but do NOT replace the seven manual playground experiments, which require genuine screenshots, request IDs, and live output captures.

7. Tripartite Root-Cause Classification

To uphold scientific integrity, we separate established findings, inferred hypotheses, and unresolved questions:
1. **Established Facts:** The 15-recording sample statistics (mean 64.45% DER, median 86.02%); the five worst rows (DER 97-100%, coverage 0-9.8%); the two empty Nova-3 RTTMs (D20260825_S003_012 and D20260825_S003_015); component identity matching DER within 0.01%; and the workbook settings re-runs.
2. **Inferred with High Confidence:** The severe DER is overwhelmingly driven by missing speech output rather than speaker attribution swapping, because missed speech accounts for >93% of error and Silero VAD detects >91% speech duration on identical audio.
3. **Unresolved Mechanisms:** Whether the speech omission occurred at the Deepgram VAD front-end, within the English language model decoder, or during benchmark word-to-RTTM conversion. Distinguishing these requires inspecting raw API JSON responses.

8. Root Cause Synthesis & Confirmation Testing Framework

Output Omission vs Converter Loss: Zero-word returns in the workbook settings sheet strongly support service-side omission. However, word-to-RTTM conversion loss remains plausible because Nova-3 RTTM files are derived from returned words using a 0.5 s gap joining heuristic. Raw word JSON is required to confirm whether words were emitted by Deepgram but dropped by the converter.
Conditional Diagnostic: diarize_model and metadata.diarize_info: Current Deepgram documentation introduces diarize_model=latest and returns diarizer build information in metadata.diarize_info. While populated workbook rows report Diarizer reported = v2, the workbook lacks raw response metadata. Thus, metadata.diarize_info must be audited during live replays, but is NOT asserted as an established cause.
Benchmark Reproducibility Caveat: Exact benchmark DER cannot be recomputed from supplied RTTMs because the benchmark author utilized an earlier reference build. File syntax, timing validity, and RTTM triplet pairings are confirmed, but exact score equality is impossible from the current files alone.
Status of Seven Manual Playground Experiments: As documented in the execution log, all seven runs remain **NOT PERFORMED - manual run required**. No manual findings are fabricated.

9. Recommendations for Deepgram Engineering & Voice Arena

- **Deepgram Engineering:** (1) Introduce an explicit 'zero-output' warning when VAD detects audio energy but ASR returns empty transcript; (2) Retain full request and response JSON payloads with request IDs in client SDKs; (3) Optimize Nova-3 acoustic front-end for far-field Indian English table-top audio; (4) Document the exact interaction between diarize_model, language, and utterance timelines.
- **Voice Arena Benchmark Team:** (1) Publish cryptographic SHA-256 hashes for all input audio and reference RTTMs; (2) Disclose exact word-to-RTTM conversion source code; (3) Report per-session speech coverage alongside DER to detect silence-collapse; (4) Retain raw API responses to distinguish service drops from evaluation pipeline errors.

10. Limitations & Retrospective

Limitations: Assessment PDF absent from workspace; workbook contains 15 rows; RTTM package contains 20 IDs; current live leaderboard values not machine-readable in local environment; manual playground runs require physical execution. **Retrospective:** The most striking finding is the bimodal performance gap: Deepgram Nova-3 functions normally on clean online audio (26.8% DER) but collapses completely on multi-party in-person Indian English audio. Given additional time, research should prioritize raw API JSON payload logging and station-level acoustic replication.